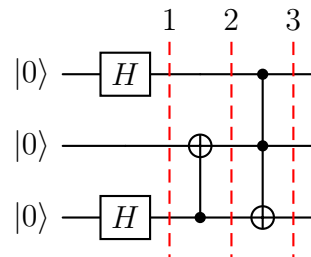


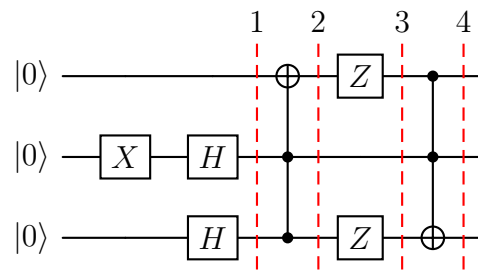
Sample of the 2nd IQC test

(This is just a trial test.)

1. Write down the state of the quantum circuits in the marked lines.

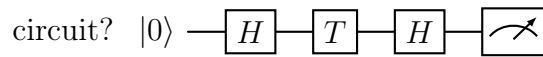


2. Write down the state of the quantum circuits in the marked lines.



3. Create an orthonormal basis set to the computational basis $|0\rangle, |1\rangle$. Orthonormal means, that the basis vectors are normalized and orthogonal to each other (*jsou normované a navzájem kolmé*). How can you check the upper define orthogonality? How can we change from computational basis to your basis?

4. What is the probability of measuring 0 as the outcome of the following



5. Explain the equal superposition (*rovnoměrná superpozice*). What is it good for?

6. Write down the state of the quantum circuits in the marked lines. Whats the probabilities of measuring the $|0\rangle$ and $|1\rangle$ in the qubits marked with the measurement symbol?

